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Para Int

Likelihood Equivalence

If $x, y \in \mathcal{X}$ are called likelihood equivalent " $\sim$ " if
 $L(\theta|x) \propto L(\theta|y)$ [i.e. if $\frac{L(\theta|x)}{L(\theta|y)} = K(x, y)$ is free of θ .]

" $\sim$ " is an equivalence relation that partitions $\mathcal{X}$ into disjoint classes (called equivalent classes). Will show: " $\sim$ " induces the minimal sufficient partition.

Define: Any stat $T(\underline{X})$ induces a partition of $\mathcal{X}$. Let $R(T)$ be the range of T .

Define $A_t = \{x \in \mathcal{X} : T(x) = t\}$

The partition induced by T is $\mathcal{U} = \{A_t : t \in R(T)\}$

$\underline{X} = (X_1, \dots, X_n)$ sample from pdf/pmf $f(x|\theta), \theta \in \Theta$
Def: Suppose $\mathcal{U}$ is a partition of $\mathcal{X}$ induced by a statistic $T(\underline{X})$. $\mathcal{U}$ is called a sufficient partition for θ if (the conditional distribution of $\underline{X}$ given $T(\underline{X}) = t$ is free of θ)
 $T(\underline{X})$ is sufficient stat. for θ

Note: (1) Given any stat T , we have a partition induced by T and given any partition $\mathcal{U}$, we can define (may not be uniquely) a stat that induces $\mathcal{U}$.

(2) If T_1, T_2 give same sufficient partition, then $T_1 = \phi(T_2)$ for some one-one ϕ .

Defⁿ: (1) Consider two partitions $\mathcal{U}_1$ and $\mathcal{U}_2$ of $\mathcal{X}$. The partition $\mathcal{U}_1$ is called a "subpartition" of $\mathcal{U}_2$ if each partition set of $\mathcal{U}_2$ is union of some partition sets of $\mathcal{U}_1$.

(2) A partition $\mathcal{U}$ (of $\mathcal{X}$) is called minimal sufficient if

- (i) $\mathcal{U}$ is a sufficient partition &
- (ii) If $\mathcal{C}$ is a sufficient partition of $\mathcal{X}$ then $\mathcal{C}$ must be a subpartition of $\mathcal{U}$.

Thm (Lehmann-Scheffé)

The likelihood equivalence " $\sim$ " induces a minimal sufficient partition of $\mathcal{X}$.

Pf: Let's check (ii) first. Start with a sufficient partition $\mathcal{C}$ of $\mathcal{X}$. Consider a statistic $S(\mathbf{x})$ that induces $\mathcal{C}$.

Enough to show, if $S(\underline{x}) = S(\underline{y}) \Rightarrow \underline{x} \sim \underline{y}$

$$\frac{f(\underline{x}|\theta)}{f(\underline{y}|\theta)} \stackrel{\text{NFFT}}{=} \frac{f_0(S(\underline{x}))h(\underline{x})}{f_0(S(\underline{y}))h(\underline{y})} = \frac{h(\underline{x})}{h(\underline{y})} \Rightarrow \underline{x} \sim \underline{y}$$

Now check (i)

Check that partition $\mathcal{U}$ is sufficient (given: $\mathcal{U}$ is obtained by " $\sim$ "). Suppose statistic $T(\mathbf{x})$ induces partition $\mathcal{U}$. Enough to show, $T(\mathbf{x})$ is sufficient.

$$A_t = \{\underline{x} \in \mathcal{X} : T(\underline{x}) = t\}. \text{ Given } \underline{x}, \underline{y} \in A_t \text{ (i.e. } \underline{x} \sim \underline{y})$$

$$f(y|\theta) = K(x, y) f(x|\theta)$$

To show that $f_{X|T}(x|t, \theta)$ is free of θ .

(Discrete X)

$$\begin{aligned} & \parallel \\ & P_\theta(X=x | T(X)=t) \\ &= \frac{P_\theta(X=x, T(X)=t)}{P_\theta(T(X)=t)} \quad (x \in A_t) \\ &= \frac{P_\theta(X=x)}{\sum_{y: T(y)=t} P_\theta(X=y)} = \frac{f(x|\theta)}{\sum_{y: T(y)=t} K(x, y) f(x|\theta)} \rightarrow \text{free of } \theta \end{aligned}$$

Alt. Def.: A statistic $T(x)$ is minimal sufficient iff it induces the minimal sufficient partition

Easier way to put the above thm:

Thm: $T(x)$ is a minimal sufficient stat for θ if the following holds

$$\forall x, y \in \mathcal{X}, \quad \underbrace{\frac{L(\theta|x)}{L(\theta|y)}}_{\text{free of } \theta} \Leftrightarrow T(x) = T(y)$$

Ex: $X_1, \dots, X_n \sim \text{Unif}(0, \theta)$. Min suff stat for $\theta = ?$

$$L(\theta|x) = \frac{1}{\theta^n} \mathbb{1}(0 \leq x_{(n)} \leq \theta). \quad T(x) = X_{(n)} \text{ is suff by NFFT}$$

$$L(\theta|y) = \frac{1}{\theta^n} \mathbb{1}(0 \leq y_{(n)} \leq \theta)$$

$$L(\theta|x) = \frac{1}{\theta^n} \mathbb{1}(0 \leq x_{(n)} \leq \theta)$$

$$= K(x, y) \frac{1}{\theta^n} \mathbb{1}(0 \leq y_{(n)} \leq \theta) \quad \forall \theta$$

$$\Rightarrow x_{(n)} = y_{(n)} \text{ b/c.}$$

if not suppose $x_{(n)} < y_{(n)}$ then take $\theta \in (x_{(n)}, y_{(n)})$

then RHS = 0 but LHS = 1

$\therefore T(X) = X_{(n)}$ is min suff for θ .

Ex. $X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, \sigma^2)$

Check it $T(X) = (\bar{X}, S^2)$ is min suff.

Ex. $X_1, \dots, X_n \stackrel{iid}{\sim} U(-\theta, \theta)$ Min suff stat
 $|X_1|, \dots, |X_n| \stackrel{iid}{\sim} U(0, \theta)$ $\left(T(X) = (X_{(1)}, X_{(n)}) \right)$
 $\theta > 0$ $\left(\max_{1 \leq i \leq n} |X_i| \right)$

Ex. Let $X_1, \dots, X_n \stackrel{iid}{\sim} \text{logistic}(\theta, 1)$ having pdf
 $f(x|\theta) = \frac{e^{-(x-\theta)}}{\{1 + e^{-(x-\theta)}\}^2}$

Show $T(X) = (X_{(1)}, \dots, X_{(n)})$ is minimal sufficient.

$$\text{pf: } x, y \in \mathcal{X} \quad \frac{f(x|\theta)}{f(y|\theta)} = \frac{\prod_{i=1}^n \frac{e^{-(x_i-\theta)}}{(1 + e^{-(x_i-\theta)})^2}}{\prod_{i=1}^n \frac{e^{-(y_i-\theta)}}{(1 + e^{-(y_i-\theta)})^2}} = \frac{e^{-\sum x_i}}{e^{-\sum y_i}} \frac{\prod_{i=1}^n (1 + e^{-(y_i-\theta)})^2}{\prod_{i=1}^n (1 + e^{-(x_i-\theta)})^2}$$

$\rightarrow$ free of θ ?
 (will come back)

Result: Let $\underline{X} = (X_1, \dots, X_n)$ be a r.v. having pmf/pdf $f(x|\theta) : \theta \in \{\theta_0, \theta_1, \dots, \theta_k\} = \Theta_0$
 $\mathcal{P}_0 = \{f(x|\theta) : \theta \in \Theta_0\}$. Show that $T(\underline{x}) = \left(\frac{f(x|\theta_1)}{f(x|\theta_0)}, \dots, \frac{f(x|\theta_k)}{f(x|\theta_0)} \right)$ is a minimal sufficient stat for $\mathcal{P}_0$

Pf: HW

Result: Let $\mathcal{P} = \{f(x|\theta) : \theta \in \Theta\}$ $\mathcal{P}_0 = \{f(x|\theta) : \theta \in \Theta_0\}$ where $\Theta_0 \subseteq \Theta$
 If T is a minimal sufficient statistic for $\mathcal{P}_0$ and sufficient stat for $\mathcal{P}$. Then T is minimal suff for $\mathcal{P}$.

Pf: Let S suff stat for $\mathcal{P}$
 $\Rightarrow S$ suff stat for $\mathcal{P}_0$
 $\Rightarrow T = h(S)$ for some fcn h ($\because T$ is min suff)

Consider $\theta_0 = 0, \theta_1, \dots, \theta_k$ fixed real no: $\theta_i \neq \theta_j$
 the finite family $\{f(x|\theta) : \theta \in \{\theta_0, \dots, \theta_k\}\}$
 $T(\underline{x}) = \left(\frac{f(x|\theta_1)}{f(x|\theta_0)}, \dots, \frac{f(x|\theta_k)}{f(x|\theta_0)} \right)$ is min suff for $\mathcal{P}_0$

$$T_j(x) = \frac{\prod_i e^{-(x_i - \theta_j)}}{\prod_i (1 + e^{-(x_i - \theta_j)})^2} \frac{\prod_i (1 + e^{-x_i})^2}{\prod_i e^{-x_i}}$$

$$= e^{n\theta_j} \prod_{i=1}^n \left[\frac{1 + e^{-x_i}}{1 + e^{-(x_i - \theta_j)}} \right]^2$$

Will show for $k = n+1$ $T(x) = T(y) \Leftrightarrow (x_{(1)}, \dots, x_{(n)})$
 $= (y_{(1)}, \dots, y_{(n)})$

$$T_j(x) = T_j(y) \quad \forall j = 1, \dots, k \quad (\text{enough to show})$$

$$\prod_1^n \left[\frac{1 + e^{-(x_i - \theta_j)}}{1 + e^{-x_i}} \right] = \prod_1^n \left[\frac{1 + e^{-(y_i - \theta_j)}}{1 + e^{-y_i}} \right]$$

$$\text{Take } e^{-x_i} = u_i, e^{-y_i} = v_i, e^{\theta_j} = z_j \quad j = 0, 1, \dots, k$$

$$T(x) = T(y)$$

$$\Rightarrow \prod_1^n \frac{1 + z u_i}{1 + u_i} = \prod_1^n \frac{1 + z v_i}{1 + v_i} \quad \forall z = z_0, z_1, \dots, z_{n+1}$$

$$\Rightarrow P_1(z) - P_2(z) = 0 \quad \forall z = z_0, z_1, \dots, z_{n+1}$$

$$\therefore P_1(z) = P_2(z) \quad \forall z > 0.$$

$$\prod_{i=1}^n (1 + z u_i) = \frac{\prod (1 + u_i)}{\prod (1 + v_i)} \prod_{i=1}^n (1 + z v_i)$$

$$\text{coeff of } z^r \text{ for } r=0 \Rightarrow \prod (1 + u_i) = \prod (1 + v_i)$$

$$\therefore T(x) = T(y) \Rightarrow \prod_{i=1}^n (q + u_i) = \prod_{i=1}^n (q + v_i) \text{ where } q = \frac{1}{3}$$

$$\text{roots are } \{-u_1, \dots, -u_n\} = \{-v_1, \dots, -v_n\}$$

$$\therefore \{x_1, \dots, x_n\} = \{y_1, \dots, y_n\} \text{ (one-one)}$$

$$\Rightarrow \underline{(x_{(1)}, \dots, x_{(n)}) = (y_{(1)}, \dots, y_{(n)})}$$

HW 1) Show the same, for Cauchy(0,1)

$$f(x|\theta) = \frac{1}{\pi \{1 + (x-\theta)^2\}}$$

$(X_{(1)}, \dots, X_{(n)})$ is min suff

2) Double Exponential...